341154(41)

B. Pharmacy (First Semester) Examination, Nov.-Dec. 2018

(PCI Scheme)

(Pharmacy Branch)

PHARMACEUTICAL INORGANIC CHEMISTRY - Theory (BP104T)

Time Allowed : Three hours

Maximum Marks : 75

Note *: Attempt questions of all sections as directed.*

Section-A
(Objective Type Questions)

Note : *Answer all the following questions. Each question carries 1 marks.*

1. Choose the correct answer : 20×1=20

(i) If the pH of blood falls below 7·3, the condition is termed as :

(a) Acidosis

(b) Alkalosis

(c) Neutral

(d) None of above

(ii) Dental caries is a medical term of :

(a) Cleaning action

(b) Polising action

(c) Tooth decay or cavities

(d) None of above

(iii) Dilute hydrochloric acid is an :

(a) Antacid

(b) Acidifying agent

(c) Cathartics

(d) Antimicrobial agent

(iv) $NaHCO_3$ is the chemical formula for :

(a) Sodium phosphate

(b) Sodium carbonate

(c) Sodium bicarbonate

(d) Sodium orthophosphate

(v) Drugs used to bring about defection :

(a) Emetics

(b) Cathartics

(c) Haemantinics

(d) None of above

(vi) An important antidote in the treatment of cyanide poisoning is :

(a) Atropine

(b) Ethanol

(c) Desferioxiamine

(d) Sodium thiosulphate

(vii) Zinc sulphate is also known as :

(a) Yellow vitriol

(b) Red vitriol

(c) White vitriol

(d) Blue vitriol

(viii) β-rays are :

(a) Positively charged

(b) Negatively charged

(c) Neutral

(d) None of the above

(ix) Which of the following is not a property of acids :

(a) Taste sour

(b) Turn blue litmus red

(c) React with base to form water

(d) Feel slippery on the skin

(x) KCl is indicated in the treatment of :

(a) Menieres syndrome

(b) Myasthenia gravis

(c) Antidote in digitallis intoxication

(d) All of the above

(xi) Isotopes are having :

(a) Same atomic number but different mass number

(b) Some atomic number and same mass number

(c) Atomic number increases

(d) None of the above

(xii) High serum potassium level is :

(a) Hyperkalemia

(b) Hypernatrimia

(c) Hypermangnesimia

(d) None of the above

(xiii) Opalescence/turbidity is given by the limit test of :

(a) Arsenic

(b) Iron

(c) Lead

(d) Chloride

(xiv) Inorganic compound which is used as an antimicrobial is :

(a) Zinc sulphate

(b) Boric acid

(c) Kaolin

(d) Ammonium chloride

(xv) Astringent act as :

(a) Protein precipitant

(b) Lipid precipitant

(c) Enzyme catalysis

(d) Bacterial inhibition

(xvi) Electrolyte replenisher is :

(a) Oral rehydration salt

(b) Chlorinated lime

(c) Potassium permanganate

(d) Activated charcoal

(xvii) Magnesium hydroxide is also known as :

(a) Epsom salt

(b) Milk of magnesia

(c) Clay

(d) Clay precipitated chalk.

(xviii) Impurities in the pharmaceutical preparations may be due to the following sources :

(a) Raw material

(b) Manufacturing process

(c) Chemical instability

(d) All of the above

(xix) A solution that maintains pH when acids or bases are added in it known as :

(a) Neutral solution

(b) Acidic solution

(c) Buffer

(d) Basic solution

(xx) A substance that on dissolving in water gives hydroxyl ion is called :

(a) Base

(b) Buffer

(c) Acid

(d) All of the above

Section-B

7×5=

(Short Answer Type Questions)

Note : Attempt any seven questions. Each question carries 5 marks.

2. Describe the principles and procedure involved in the limit test of iorn.

3. Describe the sources of impurities in pharmaceutical substances.

4. What are anticaries agenets? Write the role of fluoride in the treatment of dental caries.

5. Give the chemical formula, properities and medical uses of aluminium hydroxide gel.

6. Explain cathartics with examples. Write a note on saline cathartics.

7. Define antidotes. Give the mechanism of action of antidotes with examples.

8. Give an account of ferrous sulphate as haematinics.

9. Give classification and mechanism of action of antimicrobial agents.

10. What is buffer? Describe the importance of buffer in pharmaceutical system.

Section-C

2×10=20

(Long Answer Type Questions)

Note : *Attempt any two questions. Each question carries 10 marks.*

11. Write down the preparation, assay, properties and medicinal uses of calcium gluconate.

12. Define acid and base according to Arrhenius theory. Descibe the physiological acid base balance of the body.

13. What do you mean by radioactive susbstances and radioactivity? Explain the methods of measuring radioactivity.